

Closing the Poop Loop: The Creation of a Manure Market in India's Gobar Dhan Scheme

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Abstract

The spatial mismatch between manure supply from livestock and demand for crop nutrients is a market failure rooted in manure's transportation costs. This creates negative externalities such as nutrient runoff and soil degradation. Biogas technology is promoted as an innovation that lowers these costs by centralizing collection and processing manure into standardized outputs: energy and organic fertilizer (slurry). Because slurry has greater nutrient efficiency than raw manure, studying slurry use opens the possibility for welfare improvement. In this study, I evaluate India's 2018 Gobar Dhan scheme, which installs community biogas plants that purchase dung from livestock owners and distribute slurry to local farmers. Using plant registry data and water quality records from India's Central Water Commission, I exploit variation in water turbidity, a proxy for fertilizer runoff, across villages from 2019 to 2025. I find no significant change in water turbidity for the average treated village. However, heterogeneity analysis reveals that villages farther from fertilizer shops experience larger turbidity reductions following plant installation. The null average effect suggests that either plants are sited where the market failure is mild, or that India's heavy chemical fertilizer subsidies continue to distort behavior even after transaction costs are lowered. This study provides the first evaluation of a biogas policy on the basis of organic fertilizer use and highlights that institutional innovations to lower transaction costs may require complementary price reforms or careful targeting to realize potential welfare gains.

Keywords: solid waste, agricultural policy, externalities

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1. Introduction

The ecological and economic costs of manure have long presented a dual challenge in agricultural systems: it is simultaneously a critical resource for soil fertility when processed and a significant source of water and air pollution when unprocessed or mismanaged. This duality is rooted in a market failure: in most developing agricultural contexts, there exists no efficient market for manure as an agricultural input. Its low value-to-mass ratio results in prohibitive transportation costs, spatially constraining its exchange to immediate vicinities (Ashraf & Christensen, 1974). Consequently, supply (from livestock holdings) and demand (from crop-producing fields) are often geographically decoupled, leading to inefficient allocation. This leads to surplus manure becoming a waste disposal problem for livestock owners, while crop producers face nutrient deficits and over-reliance on chemical fertilizers (Ghosh, 2004). This mismatch creates substantial negative externalities, such as nutrient runoff, greenhouse gas emissions, and soil degradation, whose social costs are not borne by the producers, representing a clear economic inefficiency and welfare loss (Ribaudo et al., 2003; Innes, 2000).

Theoretically, private bargaining could resolve this externality and allocate manure efficiently (Coase, 1960). However, the barrier is precisely the prohibitively high transaction costs of coordinating exchange among numerous, dispersed smallholders (Stavins, 1995). These costs, such as costs of search, negotiation, quality verification, transport, and enforcement, prevent the spontaneous emergence of a manure market.

Biogas technology has been widely promoted as an innovation capable of lowering these transaction costs. By centralizing the collection and processing of livestock dung into biogas and slurry, it transforms a heterogeneous, bulky waste product into standardized, tradable outputs: energy and organic fertilizer. However, the economic literature evaluating biogas has been overwhelmingly dominated by an “energy demand” paradigm, assessing viability almost exclusively through the lens of fuel substitution and energy cost savings (Szarka et al., 2013; Khandker et al., 2012; Mwirigi et al., 2014).

The fertilizer co-product –the slurry – is typically treated as a costless input for self-use or a waste stream with disposal costs, its potential market value and role in farmer economics largely neglected (Lansing et al., 2008). This omission is significant because the agronomic benefits of organic fertilizers are well-established (Prashar et al., 2025), suggesting latent economic demand. Yet, the formation of a localized, coordinated market for this slurry, a problem of matching decentralized supply with heterogeneous demand under high transaction costs, remains an unexamined gap. The failure of centralized biogas programs, such as those in China, has been partly attributed to this lack of a functional slurry market, highlighting the real-world consequences of this oversight (Lu & Gao, 2021).

In 2018, the Government of India launched the Galvanizing Organic Bio-Agro Resources Dhan (Gobar Dhan) scheme, a national policy installing biogas plants to convert surplus manure into biogas and slurry. Its Community Model presents a unique experiment to overcome the manure market failure. By establishing a village-scale biogas plant that purchases unprocessed manure – in the form of cow dung – from livestock owners and distributes slurry to local farmers, it creates a centralized platform for exchange. This model explicitly targets the transaction cost barriers: it provides a collection point, a formal dung pricing mechanism, and a guaranteed local outlet for slurry.

This paper investigates whether and how this policy intervention succeeds in creating a functional local manure market. Broadly, does access to biogas slurry in a village lead to productivity and coordination gains from trade between heterogeneous households in the same village? Focusing on the intensive margin of fertilizer substitution, does access to a biogas plant incentivize farm households to substitute away from chemical fertilizer towards organic fertilizer? I study this question by first modeling household fertilizer decisions with access to a biogas plant. I then test the model’s mechanisms through a staggered difference-in-differences approach combined with propensity score matching, leveraging the cross-sectional variation of villages and temporal variation of biogas plant

construction under India's 2018 Gobar Dhan scheme. Empirically, I measure a plant's impact on water turbidity, a proxy for chemical fertilizer runoff. The results reveal no detectable average treatment effect on water quality, suggesting that the hypothesized substitution is not occurring at scale in the average village. However, heterogeneity analysis provides suggestive evidence consistent with the model's predictions: villages farther from fertilizer shops, where the transaction costs of obtaining chemical fertilizers are higher, experience larger reductions in turbidity, indicating that the substitution mechanism is operative where the price incentive is strongest. By doing so, this study moves beyond the energy-centric analysis of biogas to provide the first evaluation of a biogas policy, particularly India's Gobar Dhan, on the basis of organic fertilizer use, while highlighting the importance of local market conditions for policy effectiveness.

2. Review of Manure Market Failures and Biogas Technology

This literature review proceeds in four parts. First, I explain the basic mechanics of biogas technology to establish the material flows that underpin the economic analysis. Second, I review the literature on manure as an economic good, tracing the market failure from transportation costs to pollution externalities and the Coasean transaction cost framework. Third, I examine the existing economics literature on biogas, showing its dominant energy-centric paradigm. Fourth, I identify the neglected role of slurry as a fertilizer co-product and articulate the gap this study addresses.

2.1. Mechanics of a Biogas Plant

Before reviewing the economic literature on manure management, it is useful to understand the basic mechanical operation of a biogas plant. Figure 1 illustrates the material flows in a typical community-scale plant, which is the focus of this study.

At the center of the system is the biogas plant itself – an anaerobic digester that processes organic waste in the absence of oxygen (Weiland, 2010). The input to the plant is unprocessed livestock waste, also referred to as farmyard manure or simply dung. This

material is collected from farm households and fed into the digester, where a consortium of microorganisms breaks down the organic matter over a retention period of several weeks (Angelidaki et al., 2018).

The digestion process produces two distinct outputs. The primary output is biogas, a mixture of methane (50-70%) and carbon dioxide (30-50%), along with trace amounts of hydrogen sulfide and ammonia (Weiland, 2010). Biogas can be combusted directly for heat or electricity generation, or upgraded to biomethane for injection into natural gas grids or use as fuel (Angelidaki et al., 2018). Both heat and electricity flow back to farm households as a substitute for firewood, liquefied petroleum gas, or grid electricity.

The secondary output is slurry, also called digestate, which is the nutrient-rich residue remaining after digestion (Lansing et al., 2008). Unlike raw manure, slurry has several agronomic advantages. The anaerobic digestion process concentrates plant-available nutrients, reduces pathogen loads, lowers odor, and decreases the biochemical oxygen demand of the liquid waste (Weiland, 2010). These characteristics make slurry a more efficient and safer organic fertilizer than raw manure. This slurry flows back to farm households for application on crop fields. The key takeaway crucial to this study and later policy implications is that this plant transforms a heterogeneous, bulky, and low-value input (raw dung) into two standardized, higher-value outputs (biogas and slurry).

2.2. The Pollution Externality and Economic Welfare Loss

Early economic studies of manure treat it primarily as a farm production input with inherent logistical constraints. Ashraf & Christensen (1974) and Stonehouse & Narayanan (1984) identified its low value-to-mass ratio as the primary cause of high transportation costs that spatially constrain manure markets. This characteristic fosters localized markets or their complete absence, creating a spatial disconnect between manure supply (tied to livestock production) and crop nutrient demand. Schnitkey & Miranda (1993) formalized this from a production theory perspective, showing that the net value or disposal cost of a by-product like manure can directly affect livestock production decisions. When

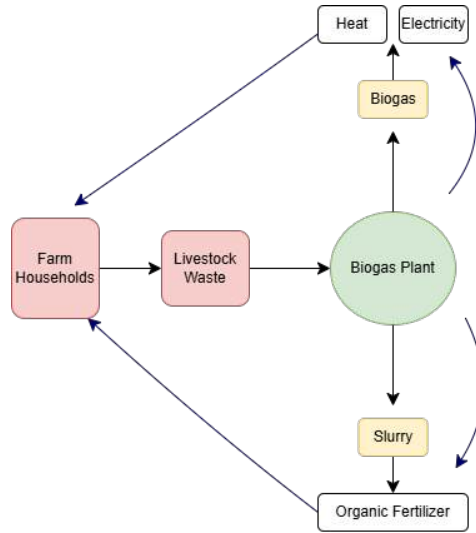


FIG. 1. Diagram of biogas plant resource flows

Note: Simplified, farm households supply unprocessed livestock waste to the biogas plant. The plant produces biogas (used for heat and electricity, returning to households) and slurry (used as organic fertilizer, returning to households). Diagram by author.

manure supply is large relative to local demand, its value falls, creating incentives for over-application or cheap disposal, despite negative externalities.

This dynamic leads directly to environmental market failure. A substantial economics literature documents the welfare losses from manure as a negative production externality. Its improper disposal, particularly nutrient runoff, imposes social costs like water and air pollution not borne by producers. Ribaud et al. (2003) demonstrate that the marginal social cost of water pollution in the form of nitrogen runoff from manure can significantly exceed its private marginal benefit for crop production, creating deadweight loss. Innes (2000) provides a formal model showing that unregulated livestock waste leads to overproduction of both livestock and pollution, with policy implications for taxes and quotas. In the Indian context, the spatial decoupling of livestock-intensive regions from crop-intensive regions exacerbates this externality, as highlighted by Ghosh (2004), who note the rising environmental cost of manure management due to changing farm structure.

The theoretical benchmark for resolving such externalities through private bargaining is the Coase Theorem (Coase, 1960), which posits that with well-defined property

rights and zero transaction costs, parties will negotiate to an efficient allocation of resources, regardless of the initial assignment of rights. This theorem has spawned a vast literature in environmental economics on the potential for market-based solutions to pollution. However, pay close attention to the theorem's limiting condition: transaction costs. Stavins (1995) rigorously analyzes how transaction costs like search, negotiation, monitoring, and enforcement costs can prevent the realization of Coasean efficiency in pollution control markets. Applied to agricultural non-point source pollution like manure runoff, these costs are typically prohibitive due to the large number of diffuse sources and affected parties, asymmetric information, and the difficulty of monitoring runoff (Horan & Shortle, 2005). This theoretical insight explains the observed market failure where the transaction costs of coordinating manure exchange between numerous smallholders are too high for a private market to emerge spontaneously.

2.3. Biogas & the Dominant "Energy Demand" Paradigm

Biogas technology is analyzed in the economics and energy literature as an institutional innovation that can lower transaction costs by creating a centralized processing point and transforming the externality (waste) into tradable private goods (energy and fertilizer). The economic viability of such plants has been predominantly evaluated through the lens of their first output: energy (biogas). The Demand-Oriented Biogas Supply (DOBS) framework, originating from analyses of the German energy transition (Szarka et al., 2013), argues that biogas production must be flexible and responsive to energy market signals (e.g. electricity grid demand) to be economically sustainable. This energy-centric focus is reflected in a large body of cost-benefit analyses and adoption studies. For instance, Khandker et al. (2012) evaluate biogas programs primarily on energy savings, time allocation (reduced fuelwood collection), and health co-benefits from reduced indoor air pollution – all linked to energy demand. Similarly, Foster and Rosenzweig (2010) model technology adoption in rural settings, focusing on learning and network effects related to energy benefits. The primary mechanism studied is the substitution

of commercial or traditional fuels, with the value of biogas often benchmarked against Liquefied Petroleum Gas (LPG) or kerosene prices (Mwirigi et al., 2014).

2.4. Neglected "Fertilizer Demand" & Local Coordination

Despite slurry being a coproduct, its role in the calculus of biogas plants is severely underexplored in the economics literature. This is a significant omission because, as Pattanayak et al. (2003) note in their study of environmental technologies, the adoption and impact of multi-output technologies depend critically on the valuation of all outputs by users. Agronomic studies consistently show the yield and soil-quality benefits of organic fertilizers (e.g. Prashar et al., 2025), implying a potential economic demand. However, the DOBS and related economic models largely treat slurry as a costless input for the plant's own use or a waste product with disposal costs (Lansing et al., 2008). The challenge of generating localized, informal demand for slurry among a heterogeneous community of smallholders – a classic problem of market formation and coordination – has received scant attention. The economics literature on rural institutions (Bardhan & Udry, 1999) and contract theory provides tools to analyze such coordination failures but has rarely been applied to manure markets. An example of the real-world consequences of this research gap is the failure of China's Centralized Biogas Plants (CBPs), documented by Lu & Gao (2021), partly attributed to this lack of a coordinated digestate market.

This research aims to fill this gap by integrating the technological insights of the biogas literature with the micro-economic models of farm-household decision-making to evaluate whether and to what extent a biogas plant creates a localized "manure market."

3. Gobar Dhan Policy

In 2018, the Government of India launched the Galvanizing Organic Bio-Agro Resources Dhan (Gobar Dhan) scheme as a flagship initiative under the Swachh Bharat Mission (Gramin) Phase II. The policy addresses three challenges: improving rural sanitation and waste management, enhancing energy security through renewable sources, and promoting

sustainable agriculture by reducing dependence on chemical fertilizers (Ministry of Drinking Water and Sanitation, 2018).

The scheme comprises four implementation models. The Individual Household Model serves single households with small-scale plants for personal use. The Cluster Model aggregates dung from multiple households to achieve economies of scale. The Community Model involves village-scale plants that process collectively supplied dung to produce biogas for community amenities and slurry for local agricultural use. The Commercial Model supports large-scale, entrepreneur-led plants producing compressed biogas for vehicle fuel. The rollout has been geographically staggered, with states like Haryana, Gujarat, Punjab, and Kerala among the early adopters. By 2023, over 5,000 community and cluster plants had been sanctioned (MNRE, 2023).

This research focuses explicitly on the Community Model, which presents a unique institutional setting to study the emergence of a localized manure market. Unlike the Individual Household Model (where production and consumption are internalized within a single unit) or the Commercial Model (which connects to external markets beyond the village), the Community Model's viability depends intrinsically on intra-village coordination between heterogeneous households. The Gram Panchayat or a village committee typically sets a price per kilogram for dung supplied to the plant (e.g. 1-2 Rs./kg), providing a direct cash incentive for livestock owners (Ghosh et al., 2020). Crucially, the slurry is usually distributed free of charge or at a nominal cost to farming households within the same village. This institutional feature is designed to stimulate local demand for organic fertilizer, but it also means the plant's financial sustainability hinges on the continued supply of dung and the perceived value of the slurry to crop producers, instead of slurry revenue like in other cases.

The Community Model thus creates a quasi-experimental setting where a new, low-transaction-cost platform for manure exchange is introduced into villages with varying farming compositions. It explicitly seeks to overcome the Coasean transaction cost barriers highlighted in Section 2 by providing (1) a centralized collection and processing

point, (2) a formalized pricing mechanism for dung, and (3) a guaranteed local outlet for slurry. However, as noted in implementation assessments, the model's success is uneven and highly dependent on local factors such as the proportion of landless livestock keepers, the presence of crop-intensive farmers, and the social capital of the village committee (Ministry of Jal Shakti, Department of Drinking Water & Sanitation, 2022). This variation across villages in both the timing of plant installation (due to staggered rollout) and the underlying farming composition provides the empirical leverage for this study.

4. Theoretical Model

The objective of this model is to show how various types of farm households respond to a fertilizer price differential with the introduction of a biogas plant in their village. Further, I assess whether scale – that is, the number and variety of households in a village – affects household responses.

4.1. Generalized Household Model

A household engages in crop and/or livestock production. This means a household produces at least one of the following outputs: crops (c), animal protein (a), and dairy (d).

Assume a household produces a total output, $Q = Q_c + Q_a + Q_d$, which is the sum of the various types of outputs described above. In this simplification, the household has 6 possible inputs, $I = \{F, S, E, L, \bar{A}, \bar{N}\}$, which include fertilizer (F), seeds (S), animal feed (E), labor (L), and fixed inputs of livestock units (A) and land (N).

Thus, the household production function is

$$Q = f(F, S, E, L, \bar{A}, \bar{N})$$

At this stage, the prices of output and variable inputs, and the quantities of fixed inputs like land and livestock units,¹ are exogenously determined. We will not be considering the circular use of inputs yet. The outputs are sold for prices $p = \{p_c, p_a, p_d\}$. The variable inputs are available at prices $w = \{w_F, w_S, w_E, w_L\}$.

The household is assumed to maximize profits by setting both output and quantities of variable inputs at optimum levels. This brings us to the cost minimization problem of the household:

$$\min_{w_F, w_S, w_E, w_L} w_F F + w_S S + w_E E + w_L L \text{ subject to } f(F, S, E, L, \bar{A}, \bar{N}) \geq q$$

The idea is that the household is trying to find the cheapest way to produce a certain output level q . The solution to this problem yields the household's optimal input demands, denoted by

$$F^*(w_F, w_S, w_E, w_L, q)$$

$$S^*(w_F, w_S, w_E, w_L, q)$$

$$E^*(w_F, w_S, w_E, w_L, q)$$

$$L^*(w_F, w_S, w_E, w_L, q)$$

¹ Livestock units will be fixed in the short run. Even if the household can acquire another livestock unit, there is a land/space constraint and livestock are often sold in their youth. It will take a long time for the new livestock unit to become productive.

The cost function is thus equal to the amount the household spends on its inputs:

$$\begin{aligned} C(w_F, w_S, w_E, w_L, q) = & w_F F(w_F, w_S, w_E, w_L, q) + \\ & w_S S(w_F, w_S, w_E, w_L, q) + \\ & w_E E(w_F, w_S, w_E, w_L, q) + \\ & w_L L(w_F, w_S, w_E, w_L, q) \end{aligned}$$

The household's profit maximization problem is

$$\max_{F, S, E, L} pf(F, S, E, L) - C(w_F, w_S, w_E, w_L, q)$$

where the household profit function is

$$\pi = pf(F, S, E, L) - C(w_F, w_S, w_E, w_L, q).$$

From the profit maximization problem, the first order conditions are

$$\frac{d\pi}{dF} = p \frac{\partial f(L, F, S, E)}{\partial F} - w_F = 0 \implies \frac{\partial f(L, F, S, E)}{\partial F} = \frac{w_F}{p}$$

$$\frac{d\pi}{dS} = p \frac{\partial f(L, F, S, E)}{\partial S} - w_S = 0 \implies \frac{\partial f(L, F, S, E)}{\partial S} = \frac{w_S}{p}$$

$$\frac{d\pi}{dE} = p \frac{\partial f(L, F, S, E)}{\partial E} - w_E = 0 \implies \frac{\partial f(L, F, S, E)}{\partial E} = \frac{w_E}{p}$$

$$\frac{d\pi}{dL} = p \frac{\partial f(F, S, E, L)}{\partial L} - w_L = 0 \implies \frac{\partial f(L, F, S, E)}{\partial L} = \frac{w_L}{p}$$

Together these four conditions define the optimal input demands. These are local maxima due to the diminishing marginal returns. We can then derive the optimal output

$$q^*(p, w_F, w_S, w_E, w_L) = f(F^*, S^*, E^*, L^*).$$

To understand how households adjust their input mix, note that if the quantity of a variable input like fertilizer changes, all else held constant, the other variable inputs would be expected to change to restore the marginal productivity conditions shown above.

4.1.1. Fertilizer Variations

Recall that the objective of this model is to see how households make fertilizer choices. I want to show that the household fertilizer portfolio depends on the price ratio of the various fertilizer types. In this section, I will introduce variations in the fertilizer quality available to the household – and by extension, variations in fertilizer prices.

First, I will formally define a fertilizer bundle F as a function of 3 components: unprocessed dung (δ), biogas slurry (ζ) and chemical fertilizer (ϕ). Each of these has a different "fertilizing efficiency" or quality. I will model this with a Constant Elasticity of Substitution (CES) function, which allows the fertilizer inputs to be imperfect substitutes.

The household fertilizer production function is

$$F(\delta, \zeta, \phi) = (\alpha_\delta \delta^\rho + \alpha_\zeta \zeta^\rho + \alpha_\phi \phi^\rho)^{1/\rho}$$

where $\alpha_\delta, \alpha_\zeta, \alpha_\phi > 0$ are the distribution parameters. The distribution parameters $\alpha_\delta, \alpha_\zeta, \alpha_\phi > 0$ represent the relative nutrient efficiency of each fertilizer type. Slurry has the highest nutrient concentration, followed by unprocessed dung, and then chemical fertilizer. Note that this greater efficiency of slurry is the crucial element of this model, opening the possibility for welfare improvement via enabling greater slurry production and use. Substitution parameter ρ determines the elasticity of substitution (σ) where $\sigma = 1/(1 - \rho)$. Imperfect substitution suggests that $0 < \rho < 1$. The village biogas plant transforms unprocessed dung into slurry, effectively converting a lower-efficiency input into a higher-efficiency one.

The intent of this fertilizer production function is to show that there are different ways to produce fertilizer/nutrients for crops. Specifically, a household can produce the same level of crop nutrients by optimizing the level of "efficiency" in its fertilizer inputs.

Household dung production is a fixed proportion of its livestock units, $\delta_{prod} = \lambda \bar{A}$. Unprocessed dung for direct use as manure is δ_{use} . Unprocessed dung sold to the biogas plant is δ_{sell} .

This creates the constraint

$$\delta_{prod} = \delta_{use} + \delta_{sell}.$$

Thus, I amend the fertilizer production function to:

$$F(\delta, \zeta, \phi) = (\alpha_{\delta} \delta_{use}^{\rho} + \alpha_{\zeta} \zeta^{\rho} + \alpha_{\phi} \phi^{\rho})^{1/\rho}$$

subject to $\delta_{prod} = \delta_{use} + \delta_{sell}$.

Recall the generalized household production function $Q = f(F, S, E, L, \bar{A}, \bar{N})$. With this fertilizer bundle, the production function becomes:

$$Q = f(F(\phi, \delta, \zeta), S, E, L, \bar{A}, \bar{N})$$

or equivalently,

$$Q = f((\alpha_{\phi} \phi^{\rho} + \alpha_{\delta} \delta_{use}^{\rho} + \alpha_{\zeta} \zeta^{\rho})^{1/\rho}, S, E, L, \bar{A}, \bar{N})$$

The prices associated with each fertilizer type are as follows. The household buys chemical fertilizer (ϕ) at a market price w_{ϕ} . The household produces unprocessed dung (δ) at price w_{δ} capturing the labor cost of dung collection. The household buys biogas slurry (ζ) at price w_{ζ} . If the household has access to a biogas plant, $w_{\zeta} > 0$, otherwise

$w_\zeta \rightarrow \infty$. The household sells a portion of its unprocessed dung (δ) to its biogas plant at price $w_\delta \zeta$.

The net cost of the fertilizer bundle is then

$$C_F^{net} = w_\delta \delta_{use} + w_\zeta \zeta + w_\phi \phi - w_\delta \zeta \delta_{sell}$$

subject to $\delta_{prod} = \delta_{use} + \delta_{sell}$.

The solution to this gives the input demands for each fertilizer type (e.g. the input demand for slurry $\zeta^*(w_\delta, w_\zeta, w_\phi, w_\delta \zeta, F)$).

4.1.2. Construction of Fertilizer Price Differentials

With the variations in fertilizer quality, the household solves the following profit-maximization problem.

$$\max_{\phi, \delta_{use}, \delta_{sell}, \zeta, S, E, L} pf(F, S, E, L, \bar{A}, \bar{N}) - w_S S - w_E E - w_L L - C_F^{net}$$

subject to:

$$F = (\alpha_\delta \delta_{use}^\rho + \alpha_\zeta \zeta^\rho + \alpha_\phi \phi^\rho)^{1/\rho}$$

$$\delta_{use} = \delta_{prod} - \delta_{sell}$$

$$\delta_{sell} \geq 0, \delta_{use} \geq 0, \zeta \geq 0, \phi \geq 0$$

The first order conditions are:

$$\frac{\partial f}{\partial F} \cdot \frac{\partial F}{\partial \delta_{use}} = w_\delta - w_\delta \zeta$$

$$\frac{\partial f}{\partial F} \cdot \frac{\partial F}{\partial \zeta} = w_\zeta$$

$$\frac{\partial f}{\partial F} \cdot \frac{\partial F}{\partial \phi} = w_{\phi}$$

Interestingly, the first of these reveals that the household sells dung to the biogas plant until the marginal value of dung equals the net cost of using it (collection cost minus sales price).

The key insight about the effect of a biogas plant comes from comparing these first order conditions. From the first two conditions and the CES function, the optimal household fertilizer portfolio satisfies:

$$\frac{\alpha_{\delta} \delta_{use}^{\rho-1}}{\alpha_{\zeta} \zeta^{\rho-1}} = \frac{w_{\delta} - w_{\delta\zeta}}{w_{\zeta}}$$

The left side of this equation is the technical substitution rate between dung and slurry. The right side is the net price differential between using dung vs. buying slurry. This equation motivates my focus on scale. The cost of using unprocessed dung as opposed to biogas slurry is the price ratio of dung *not* sold to the biogas plant and slurry.

4.1.3. Characterizing Household Types

Based on the first order conditions, I derive the following household types.

Type 1: Net Sellers of Dung Type 1 households sell most of their dung ($\delta_{sell} \approx \delta_{prod}$). These are likely smaller farms, farms specialized towards livestock production (with little crop nutrient demand), and/or labor-constrained households. Using the first order condition for dung, they satisfy

$$w_{\delta\zeta} > \frac{\partial f}{\partial F} \cdot \frac{\partial F}{\partial \delta_{use}} + w_{\delta},$$

where the revenue of selling dung to the plant exceeds the marginal value of using unprocessed dung plus the cost of dung collection.

Type 2: Balanced Fertilizer Users Type 2 households use some dung, sell some dung, and possibly buy some slurry. These are likely medium-sized farms with flexible

production. They have an interior solution where all first order conditions hold with equality.

Type 3: Net Buyers of Slurry Type 3 households may buy slurry beyond what they could produce from their own dung. These are likely crop-intensive farms with few livestock. They have a high slurry efficiency α_ζ and/or low slurry cost w_ζ .

In the absence of a biogas plant, $w_{\delta\zeta} = 0$ as there is no market for dung, and $w_\zeta \rightarrow \infty$ as slurry is unavailable.

With access to a biogas plant, $w_{\delta\zeta} > 0$ as dung-producing households obtain a new revenue stream, and $w_\zeta < \infty$ as a new input is available.

4.2. Two-Household Model Scenarios

The single-household model establishes how a household's fertilizer portfolio adjusts to the introduction of a biogas plant, which creates markets for organic options like dung and slurry. In order to model gains from trade, I now turn to consider more than one household.

Suppose there are two households who both have access to a single, centralized biogas plant. The households sell dung to the plant at price $w_{\delta\zeta}$ and buy processed slurry at price w_ζ . Each household is modeled as developed in the previous section.

The key variables for each household i are as follows. Livestock units $\bar{A}_i$ are fixed in the short run, yielding dung production $\delta_{\text{prod},i} = \lambda \bar{A}_i$. This dung is allocated between on-farm use $\delta_{\text{use},i}$ and sale to the plant $\delta_{\text{sell},i}$, subject to the constraint $\delta_{\text{prod},i} = \delta_{\text{use},i} + \delta_{\text{sell},i}$. The household may also purchase slurry ζ_i from the plant and chemical fertilizer ϕ_i from the market. These are combined into a composite fertilizer input via a Constant Elasticity of Substitution (CES) function, $F_i = (\alpha_\phi \phi_i^\rho + \alpha_\delta \delta_{\text{use},i}^\rho + \alpha_\zeta \zeta_i^\rho)^{1/\rho}$, where $\alpha_\zeta > \alpha_\delta$ reflects the higher nutrient efficiency of slurry.

Crop output for household i is given by $Q_{c,i} = f(L_i, F_i, S_i, E_i, \bar{N}_i, \bar{A}_i)$. A household is defined as crop-specific if $\bar{A}_i = 0$ and $\bar{N}_i > 0$, and livestock-specific if $\bar{N}_i = 0$ and $\bar{A}_i > 0$.

4.2.1. Scenario 1: Two Identical Mixed Farms

In this scenario, both households are identical mixed farms with $\bar{A}_1 = \bar{A}_2 > 0$ and $\bar{N}_1 = \bar{N}_2 > 0$. Prior to the biogas plant's introduction, with $w_{\delta\zeta} = 0$ and $w_\zeta \rightarrow \infty$, each household's optimal decision is to use all its dung on-farm ($\delta_{\text{use},i} = \delta_{\text{prod},i}$) and supplement any nutrient deficit with chemical fertilizer ϕ_i .

The introduction of the biogas plant alters the household's decision by creating a positive price for dung, $w_{\delta\zeta} > 0$, and a finite price for slurry, $w_\zeta < \infty$. The first order condition for dung use becomes

$$\frac{\partial f}{\partial F} \cdot \frac{\partial F}{\partial \delta_{\text{use}}} = w_\delta - w_{\delta\zeta}.$$

This condition reveals that the opportunity cost of using a unit of dung on-farm is now the foregone revenue from selling it. Consequently, each household will find it optimal to sell a portion of its dung ($\delta_{\text{sell},i} > 0$) and reduce its direct dung use. The revenue from dung sales can then be used to purchase slurry, guided by the optimality condition comparing the marginal rate of technical substitution between dung and slurry to their net price ratio.

The biogas plant enhances individual farm productivity in this scenario through two primary mechanisms. First, it enables an *efficiency gain* by allowing households to convert low-efficiency dung (α_δ) into high-efficiency slurry (α_ζ). This means the same underlying manure resource produces a larger effective fertilizer input F_i . Second, it facilitates *input re-optimization* based on new relative prices. The plant acts as a processor that allows each farm to more efficiently allocate its own organic nutrient resources, leading to a higher composite fertilizer input for a given livestock base, thereby increasing crop productivity $Q_{c,i}$ and/or reducing reliance on chemical fertilizer ϕ_i .

4.2.2. Scenario 2: Manure Coordination Between Specialized Farms

This scenario models a village with one crop-specific household H_c ($\bar{A}_c = 0, \bar{N}_c > 0$) and one livestock-specific household H_l ($\bar{N}_l = 0, \bar{A}_l > 0$). In the pre-biogas equilibrium,

a coordination failure is evident. The crop household H_c , lacking livestock, must rely entirely on chemical fertilizer (ϕ_c). The livestock household H_l , lacking cropland, has no productive use for its dung ($\delta_{\text{use},l} = 0$); the dung represents a waste product with a potentially negative value due to disposal costs and pollution externalities.

The biogas plant resolves this failure by creating a market that enables *manure coordination*. The livestock household H_l now sells all its dung to the plant ($\delta_{\text{sell},l} = \delta_{\text{prod},l}$), as the marginal value of using it on its own non-existent land is zero. This provides H_l with a new revenue stream, turning a waste product into a source of profit. The crop household H_c , for the first time, gains access to organic fertilizer. It will purchase slurry ($\zeta_c > 0$) as long as its marginal product, given by the condition $\frac{\partial f}{\partial F} \cdot \frac{\partial F}{\partial \zeta} = w_\zeta$, justifies its cost relative to chemical fertilizer.

The plant's viability hinges on the slurry demand from H_c being sufficient to absorb the supply derived from H_l 's dung, with the price difference $w_\zeta - w_{\delta\zeta}$ covering operational costs. The productivity implications are profound. The crop household's productivity increases significantly as it substitutes chemical fertilizer with cheaper, soil-improving organic slurry, reducing costs and potentially boosting yields. The livestock household's profitability rises as manure management becomes a revenue stream. At the village level, the plant internalizes a positive externality, coordinating resources between specialized farms and increasing aggregate agricultural productivity.

4.2.3. Scenario 3: Non-Viability with Geographic Separation

The final scenario considers a livestock household H_l in Village A and a crop household H_c in an adjacent Village B, with the biogas plant located in Village A. This introduces a critical element: transaction costs TC associated with the physical distance between the slurry source and the nutrient sink. These costs include transportation, negotiation, and information search.

The effective price of slurry for the crop household H_c is now $w_\zeta + TC$. Its decision to purchase slurry is governed by $\frac{\partial f}{\partial F} \cdot \frac{\partial F}{\partial \zeta} = w_\zeta + TC$. If these transaction costs TC are

sufficiently high, this condition will not hold, and H_c will find it economically rational to rely solely on chemical fertilizer. The demand for slurry falls to zero.

With no demand for its output, the biogas plant cannot generate revenue from slurry sales. Consequently, it has no incentive to purchase dung from H_l , as it cannot profitably process and dispose of the output. This demonstrates that a village-level biogas plant is ineffective at resolving coordination failures when slurry demand is geographically separated from supply by high transaction costs. However, a larger, regional plant that aggregates dung from multiple villages and distributes slurry over a wider area could potentially overcome this separation by achieving the scale necessary to absorb transport costs, though such a model would require a different institutional design than the Gobar Dhan community plant.

4.3. *Model Takeaways*

The model first establishes that a biogas plant has the *potential* to create significant value by converting a waste product (dung) into a high-efficiency fertilizer (slurry), enabling both efficiency gains for mixed farms and gains-from-trade between specialized farms. The extension from a single-household to a two-household model reveals that a biogas plant creates a market for organic fertilizer that can improve efficiency for mixed farms, enable productivity-enhancing coordination between specialized farms, or fail entirely when local demand is absent. This theoretical framework provides strong justification for hypothesizing that the success of schemes like Gobar Dhan will depend critically on local farming composition and the transaction costs of trading slurry.

5. **Data & Exploratory Analysis**

The theoretical model in Section 4 establishes that a biogas plant can create welfare gains by transforming low-efficiency dung into high-efficiency slurry, enabling both efficiency gains for mixed farms and gains-from-trade between specialized farms. However, the model also predicts that these gains depend on the relative price of chemical versus

organic fertilizers and on the transaction costs of coordinating exchange. This study empirically evaluates whether the Gobar Dhan scheme's community biogas plants succeed in inducing fertilizer substitution in practice. Testing this hypothesis requires integrating multiple data sources to construct a comprehensive panel of village-level characteristics, treatment timing, and environmental outcomes. This study constructs a sample from over 360,000 villages (1,347 treated) in 22 states, restricting to villages within a 7 km radius of a treated village that contain at least one water monitoring station within their administrative boundary. The final sample comprises 5,180 villages (239 treated) across 210 districts and 9 climate zones.

5.1. *Data Sources*

This analysis draws on multiple datasets compiled from the Socioeconomic High-Resolution Rural-Urban Geographic Dataset on India (SHRUG) v2.1 database (Asher et al., 2019) and India's Central Water Commission (CWC) to examine organic substitution across villages in India given their participation in the Gobar Dhan Scheme. Given the lack of village-level fertilizer use data after 2018, around the time of this scheme, I proxy village fertilizer use with fertilizer runoff. I measure the monthly average turbidity (in Nephelometric Turbidity Units or NTUs) of surface water at stations within the village. According to the 2011 Population Census, the average size of an Indian village is about 3.5 km², which the literature shows is sufficient distance for fertilizer runoff to be detected. This outcome variable relies on the CWC's Surface Water Quality report, which uses 809 water stations monitored monthly between 2000 and 2025, with an average of 405 stations reporting per year.

I assign a village's treatment according to the Gobar Dhan Plant Registry, an administrative dataset from the Ministry of New and Renewable Energy (MNRE) that records all sanctioned biogas projects from 2019 to 2025, including project type, status (e.g. functional, defunct), technical parameters (e.g. slurry output), capital expenditure, and precise geographic identifiers down to the coordinate level. I use this to identify

villages that received a functional community biogas plant, and the year and month construction was completed.

For the determinants of slurry uptake, I organize variables into demand-side and supply-side factors. On the demand side, I incorporate village-level data from the 2011 Socio-Economic and Caste Census (SECC), which includes information on social group composition, land ownership, share of income from cultivation, and household ownership of agriculture-related equipment. These variables capture variation in economic preferences and access to agricultural assets. Additionally, I use the Mission Antyodaya Survey (MAS) village-level dataset from 2017 to 2022, which records indicators of market and service access, such as distance to fertilizer or seed markets, and rural development indicators such as the village's Open Defecation Free status and presence of a waste disposal system.

On the supply side, I incorporate variables that capture biomass availability and topographic constraints. Forest cover data from the Vegetation Continuous Fields (VCF) dataset (2011-2020) proxy the availability of biomass fuel sources like firewood, which may influence biogas demand. Additionally, I account for geographical constraints using the World Bank's Climate Change Knowledge Portal and its Koppen-Geiger Climate Classification of Indian states. States are classified according to ten climate zones (e.g. Tropical Wet-Dry or Humid), soil type (e.g. red soil or laterite soil), annual precipitation (in mm), irrigation coverage percentage, cropping intensity percentage, and water stress index (e.g. Low, Medium, High).

5.2. Exploratory Analysis

5.2.1. Geography of Treatment and Monitoring

Figure 2 illustrates the geographic distribution of Gobar Dhan plants and Central Water Commission (CWC) monitoring stations across Indian states. The map reveals substantial geographic variability in plant rollout, with treatment villages spanning nearly all of India's climate zones, with the notable exception of cold desert regions. This absence

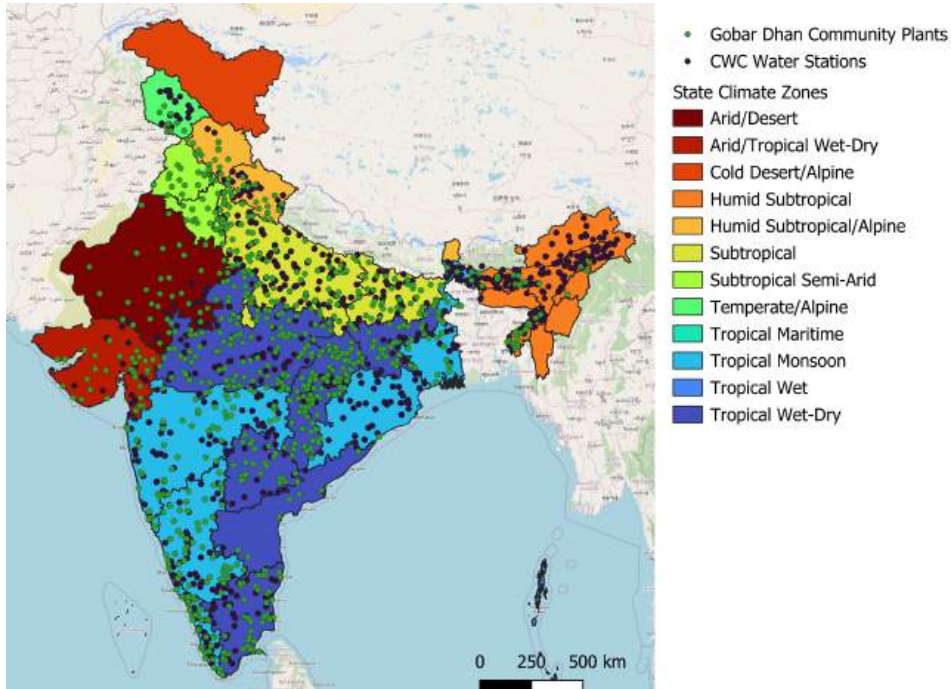


FIG. 2. Map of Gobar Dhan Plants and CWC Water Stations Across States

is consistent with the technical requirements of biogas digestion, which necessitate temperate conditions for optimal microbial activity and gas production. The spatial distribution suggests that the scheme has achieved broad geographic coverage. The map also highlights the spatial relationship between treatment villages and water quality monitoring infrastructure. The CWC stations, while concentrated in certain regions, provide reasonable coverage across the major river basins where fertilizer runoff is most consequential for water quality outcomes. This coverage enables the empirical strategy of using turbidity as a proxy for chemical fertilizer use.

5.2.2. Timing of Treatment

Figure 3 displays the distribution of treatment timing across villages in the matched sample. The sample includes 425 never-treated control villages and 239 treated villages, achieving a roughly 1.8:1 control-to-treatment ratio after matching. Treatment adoption is concentrated in the middle years of the program: 2022 (72 villages) and 2023 (92 villages)

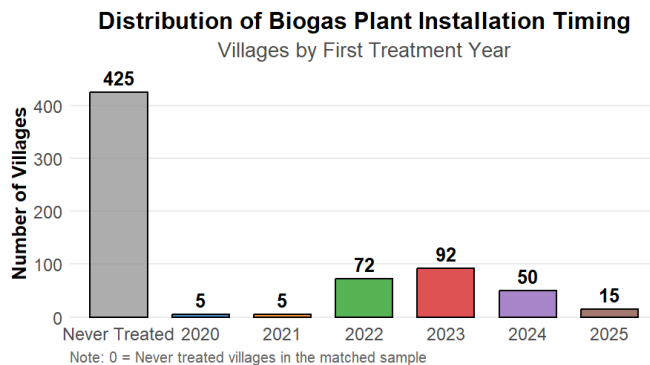


FIG. 3. Distribution of Biogas Plant Installation Timing

Notes: The figure shows the distribution of first treatment years for villages in the propensity-score matched sample. Never-treated control villages (N=425) are shown in gray, while treated villages (N=239) are distributed across 2020-2025. The majority of plants were installed in 2022-2023, with very few installations in the early years of the program.

account for over 68% of treated villages. Very few plants were installed in 2020-2021 (only 10 villages total), and the sample includes 15 villages treated in 2024 and 2025. This concentration of treatment in the middle years has implications for the staggered difference-in-differences design, as there are limited early-treated units that could serve as controls for later cohorts.

5.2.3. Characteristics of Treated and Untreated Villages

Table 1 presents summary statistics comparing treated and control villages across key covariates that theory suggests may influence both selection into treatment and outcomes of interest. As established in the theoretical model, physical infrastructure such as irrigation, farm equipment, and roads play an important role in determining a village's demand for a biogas plant.

To construct a credible control group for estimating the causal effects of biogas plants on village-level outcomes, I employed a geographic ring sampling (or "donut-hole") approach in QGIS. This spatial sampling strategy was designed to balance two competing priorities: (1) ensuring comparability between treated and control villages by selecting controls from similar geographic contexts, and (2) avoiding contamination of the control

TABLE 1 *Summary Statistics: Treated vs Control Villages*

Variable	Means Treated	Means Control	Difference	P-value
HH engaged in farming %	27.864 (13.773)	46.211 (30.288)	-18.347	0.000
Village ODF Status	1.991 (0.096)	1.451 (0.498)	0.540	0.000
Mechanized equipment ownership %	3.849 (4.742)	3.762 (3.958)	0.087	0.591
Irrigation equipment ownership %	4.430 (3.134)	8.381 (7.327)	-3.952	0.000
Agricultural equipment ownership %	6.888 (5.957)	9.489 (7.830)	-2.601	0.000
Distance to local produce market (km)	3.741 (1.696)	5.246 (3.967)	-1.505	0.000
Distance to soil testing center (km)	7.530 (1.489)	9.000 (4.880)	-1.470	0.000
Distance to govt seed center (km)	7.376 (1.022)	7.852 (4.786)	-0.476	0.000
Distance to fertilizer shop (km)	7.472 (1.459)	6.499 (4.314)	0.973	0.000

Notes: Values are means with standard deviations in parentheses.

N (Treated) = 239, N (Control) = 4,941.

Distance scale midpoints: 1=0.5km, 2=1.5km, 3=3.5km, 4=7.5km, 5=15km.

Village ODF scale: 1 = yes, 2 = no.

group by potential spillover effects from nearby biogas plants. The resulting sample sizes of 4,941 control villages and 239 treated villages provide reasonable statistical power for detecting treatment effects, though the imbalance (approximately 32:1 control-to-treatment ratio) highlights the importance of further appropriate matching techniques to avoid extrapolation bias. The propensity score matching approach described in the next section aims to construct a control group that balances these observable differences and strengthens the credibility of causal inference.

Table 1 reveals that treated villages exhibit significantly lower rates of household engagement in farming. This suggests that biogas plants are located in villages with less agricultural activity, potentially indicating that these plants serve households with alternative livelihoods or that the program targeted areas with greater non-farm employment opportunities. Further, treated villages are substantially closer to markets and government seed centers, but farther from fertilizer shops. This mixed pattern is interesting: while treated villages have better access to output markets, they face greater distances to fertilizer inputs. This could affect the economics of biogas byproduct use. For instance, if fertilizer is harder to access, the value of slurry as a substitute may be higher, but the costs of transporting slurry to fields may also be greater. Treated villages also have significantly lower rates of irrigation equipment ownership and agricultural equipment ownership, suggesting less capital-intensive agriculture. However, mechanized equipment

ownership shows no significant difference. Treated villages are also substantially more likely to be Open Defecation Free, which may reflect better sanitation infrastructure or greater development orientation – factors that could correlate with both program placement and the capacity to adopt biogas technology.

The systematic differences across multiple covariates – lower farming engagement, better market access but poorer input access, lower agricultural capitalization but better sanitation – underscore the importance of the identification strategy. These patterns suggest treatment assignment is not random but correlates with observable village characteristics in complex ways.

5.2.4. Water Turbidity as a Proxy for Fertilizer Use

A central challenge of this study is the absence of village-level fertilizer use data after 2018, precisely when the Gobar Dhan scheme began rollout. To address this limitation, I propose using surface water turbidity, measured monthly at CWC monitoring stations, as a proxy for nitrogen loading from agricultural runoff. The scientific literature establishes a clear biophysical mechanism linking chemical fertilizer application to surface water turbidity. Isotopic tracing confirms that fertilizers from agricultural soil contaminate coastal waters (Sarma et al., 2025), while direct observation in a Maharashtra lake shows that nitrate concentrations peak following rainfall and correlate with chlorophyll-a content (Moon, Durge, & Wadhai, 2024). When farmers apply nitrogenous fertilizers, excess nutrients not taken up by crops are mobilized during rainfall events, particularly during India's monsoon season. These nutrients fuel algal blooms, and the subsequent microbial decomposition increases suspended particulate matter in the water column, which is directly measured as turbidity. Thus, while turbidity is not a direct measure of fertilizer application, it captures a consequential outcome – nutrient runoff and eutrophication – that is of direct policy interest.

To validate whether turbidity captures meaningful variation in fertilizer use, I regress district-level nitrogen fertilizer consumption (thousand tons) against annual average

turbidity (NTU) using district-level panel data from 2000 to 2017. Table 2 presents results from five specifications that progressively add fixed effects: (1) no fixed effects, (2) state fixed effects, (3) district fixed effects, (4) year fixed effects, and (5) district and year fixed effects. The sample is limited to 101 district-year observations, reflecting the intersection of the ICRISAT district-level fertilizer data with CWC turbidity monitoring stations that have complete annual records.

TABLE 2 *Relationship Between District-Level Nitrogen Consumption and Water Turbidity*

	Dependent Variable: Nitrogen Consumption (thousand tons)				
	(1)	(2)	(3)	(4)	(5)
Annual Avg. Turbidity	-5.2* (2.0)	-0.12 (1.4)	0.03 (0.70)	-5.0* (2.2)	-0.16 (0.83)
Constant	96.0*** (7.7)				
Fixed Effects					
State FE	No	Yes	No	No	No
District FE	No	No	Yes	No	Yes
Year FE	No	No	No	Yes	Yes
Observations	101	101	101	101	101
R ²	0.06	0.69	0.95	0.08	0.96
Within R ²	–	0.000075	0.000024	0.05	0.0005

Note: Standard errors in parentheses.

*p<0.05, ***p<0.001.

N = 101 district-year observations.

The expected mechanism is that more fertilizer application leads to greater nutrient runoff, algal blooms, and higher turbidity. The negative coefficients in specification (4) suggests that at the district-year level, confounding factors dominate. More importantly for this study's research design, specifications (2), (3), and (5) show that once state or district fixed effects are added, the coefficient shrinks to near zero and becomes statistically insignificant. The within R-sq values are extremely small, indicating that turbidity explains virtually none of the within-district variation in fertilizer use over time.

This is precisely the variation that a difference-in-differences design relies upon: changes within a village over time, not cross-sectional differences between villages.

This exploratory validation suggests two competing possibilities, which the main DiD design will attempt to adjudicate. One possibility is that district-level fixed effects absorb meaningful variation such that village-level turbidity may also fail to respond to changes in fertilizer use. This points to a failed proxy rather than a true absence of treatment effects. A second possibility is that the district-level null results reflect aggregation bias, not the absence of a relationship at the village level. District-level averages obscure the spatial proximity between fertilizer application and water monitoring stations. Annual averaging obscures the seasonal lag between application (pre-monsoon) and runoff (monsoon). The main analysis addresses both limitations by (1) using village-level treatment assignment that matches water stations to within 7 km of village boundaries, and (2) using monthly turbidity measurements with climate controls.

Thus, the exploratory analysis does not definitively validate or invalidate turbidity as a proxy. Rather, the validation exercise is limited by data availability. Reliable, pre-2018 fertilizer consumption data exist only at the district level, forcing me to aggregate turbidity to match that coarse spatial resolution. This aggregation masks the localized relationship between a village's fertilizer use and the turbidity of its downstream water body – the very relationship the main analysis seeks to exploit. The small sample size (N=101) further limits statistical power. Given these constraints, this exercise cautions that the proxy is imperfect, and any null results in the main analysis must be interpreted with this measurement error in mind.

6. Empirical Strategy

This study evaluates whether the Gobar Dhan scheme's community biogas plants succeed in incentivizing substitution away from chemical fertilizers. The policy intervention, launched in 2018 and rolled out across Indian villages between 2019 and 2025, establishes village-scale facilities that purchase unprocessed dung from livestock owners

and distribute nutrient-rich slurry to farmers. By creating a centralized platform for exchange, these plants aim to overcome the prohibitively high transaction costs that have historically prevented the emergence of spontaneous manure markets.

The central hypothesis is that access to biogas slurry reduces chemical fertilizer application, manifesting as improvements in water quality through reduced nitrogen runoff. Villages that receive biogas plants may differ systematically from those that do not in ways that also affect the outcomes of interest. The staggered rollout of plants across villages over a six-year period provides both the empirical variation and the methodological framework to overcome this challenge.

I employ a difference-in-differences strategy combined with propensity score matching (DiD-PSM) that exploits variation in both the timing of plant construction across villages and the cross-sectional variation in village characteristics. The approach proceeds in two stages. First, propensity score matching constructs a credible control group by re-weighting untreated villages to match treated villages on pre-treatment observable characteristics identified by the theoretical model and Table 1 as likely correlated with both plant placement and fertilizer substitution behavior. Second, a staggered difference-in-differences estimator compares changes in outcomes between matched treated and control villages while accounting for the staggered timing of treatment adoption and potential heterogeneity in treatment effects.

6.1. *Propensity Score Matching*

This section addresses selection on time-varying observables. The theoretical framework in Section 4 highlights specific characteristics that should correlate with both plant placement and fertilizer substitution behavior. Table 1 confirms that treated and control villages differ systematically across precisely these dimensions: treated villages have significantly lower rates of household engagement in farming, better access to markets but poorer access to fertilizer inputs, lower agricultural capitalization but better sanitation infrastructure. These patterns suggest that villages with greater cattle endowments,

different farming compositions, and varying market access may have both higher likelihood of receiving plants and different baseline fertilizer use patterns.

I estimate propensity scores using a probit model of treatment status on pre-treatment (2017) village characteristics drawn from the Mission Antyodaya surveys and Census data. The covariates include agricultural characteristics such as net sown area, irrigated area, cropping intensity, and the share of households engaged in farming. Asset ownership is captured through the shares of households owning mechanized farm equipment, irrigation equipment, and agricultural equipment. Market access is measured by distances to fertilizer shops, seed centers, soil testing centers, and local produce markets. Infrastructure indicators include Open Defecation Free status and the presence of waste disposal systems. Finally, I include climate zone and state fixed effects to absorb geographic heterogeneity.

Each treated village is matched to up to three control villages with replacement, using caliper matching to ensure covariate balance. The matched sample then serves as the basis for the staggered difference-in-differences estimations.

6.2. *Staggered Difference-in-Differences Framework*

The rollout of biogas plants under the Gobar Dhan scheme is staggered across villages between 2019 and 2025. The core of my identification strategy is a staggered DiD design that compares changes in outcomes for treated villages (those receiving a biogas plant) to changes for untreated villages, accounting for variation in treatment timing. Following recent advances in the treatment effects literature, I estimate a two-way fixed effects specification while also employing the robust estimator developed by Callaway and Sant’Anna (2021), which addresses biases that can arise when treatment effects are heterogeneous across units or over time (Goodman-Bacon, 2021).

The baseline specification takes the following form:

$$Y_{it} = \alpha_i + \lambda_t + \delta D_{it} + \mathbf{X}_{it}'\beta + \varepsilon_{it}$$

where Y_{it} is the outcome of interest (turbidity) for village i in month t ; α_i are village fixed effects, absorbing all time-invariant village characteristics; λ_t are year fixed effects, absorbing common shocks; and D_{it} is an indicator that switches to one in the periods after village i receives a biogas plant, and zero otherwise. $\mathbf{X}_{it}$ is a vector of time-varying controls, including local fertilizer market availability and climate conditions. The coefficient of interest, δ , captures the average treatment effect on the treated (ATT) under the identifying assumption of parallel trends, where in the absence of treatment, outcomes for treated and control villages would have followed parallel paths.

However, the traditional two-way fixed effects estimator has been shown to produce biased estimates when treatment effects are heterogeneous across units or over time (Goodman-Bacon, 2021). The bias emerges because already-treated units serve as controls for later-treated units, contaminating the comparison group. To address this concern, I complement the TWFE estimates with the Callaway-Sant’Anna estimator, which constructs clean comparisons by comparing each treated cohort only to units that are either never treated or not yet treated at that time. This ensures that the control group is never contaminated by already-treated units. Results from both estimators are presented, with the Callaway-Sant’Anna estimates serving as the preferred specification.

The credibility of the staggered DiD design hinges on whether outcome trends were similar between treated and control villages before plant construction. I test this by estimating event-study specifications that include leads of treatment indicators. Statistically insignificant coefficients in the pre-treatment periods provide evidence supporting the parallel trends assumption, while significant pre-trends would suggest that treatment and control villages were on different trajectories before the policy, potentially biasing the estimates.

6.3. *Heterogeneity Analysis*

The main analysis assumes homogeneous treatment effects across villages. However, the theoretical framework suggests that the impact of biogas plants on fertilizer substitution

may vary with village characteristics that affect the costs of accessing complementary inputs. I estimate heterogeneity using district, year, and month fixed effects, with standard errors clustered at the village level.

I examine three distance-based measures of market access. Distance to fertilizer shop directly proxies the transaction cost of obtaining chemical fertilizer; the theoretical mechanism predicts that villages farther from fertilizer shops should experience larger turbidity reductions (more substitution away from chemical fertilizers) in response to a biogas plant. Distance to market proxies for general economic activity and agricultural intensification, for which the predicted sign is ambiguous. Distance to seed center proxies for access to complementary agricultural inputs and general rural isolation.

The analysis proceeds using the equation:

$$\begin{aligned}
Y_{it} = & \gamma_d + \gamma_y + \gamma_m + \delta_1 \text{Treated}_{it} \\
& + \delta_2 (\text{Treated}_{it} \times \text{DistMarket}_i) \\
& + \delta_3 (\text{Treated}_{it} \times \text{DistFertShop}_i) \\
& + \delta_4 (\text{Treated}_{it} \times \text{DistSeed}_i) + \varepsilon_{it}
\end{aligned}$$

where γ_d , γ_y , and γ_m are district, year, and month fixed effects respectively. Standard errors are clustered at the village level. The coefficient δ_1 captures the treatment effect for villages at the baseline distance to all markets (the reference group), while δ_2 , δ_3 , and δ_4 capture how the treatment effect changes with each kilometer increase in distance to markets, fertilizer shops, and seed centers respectively.

7. Results

7.1. Propensity Score Matching Results

To address potential selection bias in biogas plant placement, I construct a matched control group using propensity score matching. The matching is performed on pre-treatment (2017) village characteristics that correlate with both program placement and fertilizer substitution behavior, as identified in the theoretical framework.

TABLE 3 *Covariate Balance Before and After Propensity Score Matching*

Variable	Before Matching		After Matching		Var. Ratio
	Treated Mean	Control Mean	Treated Mean	Control Mean	
Propensity Score	0.247	0.129	0.247	0.247	1.031
HH engaged in farming (%)	27.86	48.99	27.86	27.56	1.287
Distance to Market (km)	3.74	5.22	3.74	3.77	0.997
Distance to Soil Testing (km)	7.53	8.86	7.53	7.62	0.939
Distance to Seed Center (km)	7.38	7.69	7.38	7.56	0.458
Distance to Fertilizer Shop (km)	7.47	6.64	7.47	7.56	0.933
Balance Statistics					
Sample Size	4941	239	239	425	
Std. Mean Diff. (Mean)		1.057		0.058	
Std. Mean Diff. (Max)		2.159		0.182	
R ²		0.234		0.002	

Table 3 presents covariate balance before and after matching. The propensity score matching procedure achieves excellent covariate balance across all dimensions. Before matching, treated and control villages exhibited substantial and systematic differences. Treated villages had considerably lower shares of households engaged in farming (27.9% vs. 49.0%), were closer to markets (3.74km vs. 5.22km), and had different patterns of access to agricultural infrastructure. These differences produced large standardized mean differences, with the mean absolute SMD of 1.057 and a maximum of 2.159, indicating that the raw control group was not a valid counterfactual for treated villages.

After an approximate 1:1 nearest neighbor matching without replacement, all covariates are well-balanced. The mean absolute standardized mean difference falls dramatically to 0.058, well below the conventional threshold of 0.2, and the maximum SMD drops to 0.182. The variance ratios for most covariates are close to 1, indicating

similar distributions between treated and matched control groups. The R-sq from a probit regression of treatment on covariates declines from 0.234 to 0.002, implying that after matching, the covariates have essentially no power to predict treatment status.

The matching procedure preserves the full sample of 239 treated villages while identifying 425 comparable control villages from an initial pool of 77,952. This balanced sample provides a credible foundation for estimating treatment effects.

The successful balancing achieved through propensity score matching addresses a key threat to identification: selection on observables. By ensuring that treated and control villages are comparable across the dimensions most likely to correlate with both program placement and outcomes, the matching procedure reduces the risk that estimated treatment effects are confounded by systematic differences between groups.

This said, matching cannot address selection on unobservable factors that affect both program placement and outcomes but are not captured in the data. The subsequent staggered difference-in-differences design with village fixed effects helps mitigate this concern by differencing out time-invariant unobserved heterogeneity. Further, the matching procedure restricts attention to the region of common support, meaning that the estimated effects apply to villages with characteristics similar to those that actually received treatment, rather than to the entire population of Indian villages.

With a well-balanced matched sample in hand, I now turn to estimating the causal effects of biogas plant installation on water quality using the staggered difference-in-differences framework described in Section 6.2.

7.2. Staggered Difference-in-Differences Estimates

The propensity score matching procedure established that treated and matched control villages are comparable on observable pre-treatment characteristics. Under the assumption that any remaining unobserved differences are time-invariant and can be absorbed by village fixed effects, the staggered difference-in-differences estimator identifies the causal effect of biogas plant installation on water turbidity.

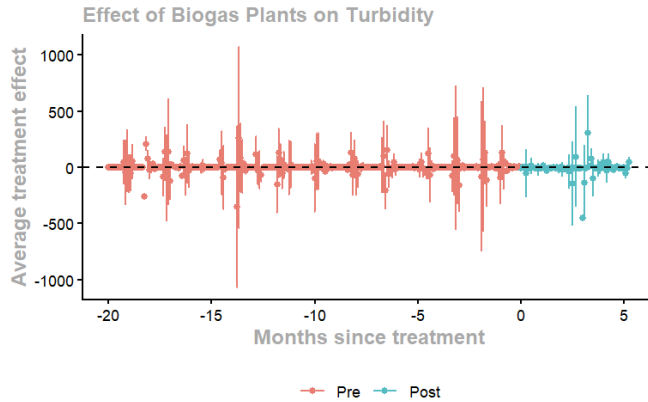


FIG. 4. Event Study: Dynamic Treatment Effects on Water Turbidity

Notes: The figure plots dynamic treatment effects from the Callaway and Sant’Anna (2021) staggered difference-in-differences estimator. Points represent estimated coefficients for each event time (months since treatment), with vertical lines indicating 95% confidence bands. Standard errors are clustered at the village level. The omitted category is the month immediately before treatment ($t = -1$). The pre-treatment period spans months -20 to -2 , though only months -5 through -2 are shown in Table 4 for brevity.

TABLE 4 *Dynamic Treatment Effects: Event Study Estimates*

Months Since Treatment	Coefficient	Std. Error	95% CI		P-value
			Lower CI	Upper CI	
-5	-8.008	3.470	-14.809	-1.208	0.021
-4	5.442	3.163	-0.758	11.641	0.085
-3	-0.300	4.683	-9.478	8.878	0.949
-2	-1.950	6.474	-14.639	10.739	0.763
0	-7.092	3.657	-14.260	0.076	0.052
1	-3.333	4.143	-11.454	4.787	0.421
2	-2.883	2.364	-7.517	1.750	0.223
3	-51.658	53.375	-156.273	52.956	0.333
4	0.658	5.819	-10.747	12.064	0.910
5	-5.092	3.790	-12.520	2.337	0.179

Note: Standard errors clustered at the village level. The omitted category is the month immediately before treatment ($t = -1$). The large standard error at $t = 3$ reflects the limited number of villages with three or more years of post-treatment data, given the concentration of plant installations in 2022-2023.

Figure 4 presents an overview of the dynamic treatment effects from the Callaway and Sant’Anna (2021) estimator. Table 4 presents a subset of point estimates closer to the treated month.² Among the four pre-treatment periods shown, only the coefficient

² The pre-treatment period spans months -20 to -2 , though only months -5 through -2 are shown in Table 4 for brevity.

at $t = -5$ is statistically significant at conventional levels, and its confidence interval is wide. The remaining pre-treatment coefficients are statistically indistinguishable from zero, with no systematic pattern of divergence prior to treatment. This pattern supports the parallel trends assumption, suggesting that treated and control villages were on similar trajectories before plant installation.

Turning to the post-treatment period, the coefficients reveal a consistent pattern of null effects. In the period of treatment ($t = 0$), the estimated effect is -7.09 NTU with a standard error of 3.66 ($p = 0.052$), narrowly missing conventional significance thresholds. In the subsequent post-treatment months, all coefficients are statistically insignificant. While we cannot rule out small or moderate effects, the confidence intervals are narrow enough that we can reject the possibility of very large effects (e.g. turbidity dropping by 100 NTU).

As seen in Figure 4, the absence of a clear treatment effect is consistent across all post-treatment periods, with no apparent trend toward either improvement or deterioration over time. This null finding suggests that, on average, biogas plant installation has not led to detectable reductions in water turbidity during the study period. If the price differential created by the plant is insufficient to overcome existing behavioral or credit constraints in the average village, a null average effect is expected. I therefore turn to a heterogeneity analysis to test whether the predicted substitution mechanism is present in villages where the incentive to substitute is strongest.

7.3. *Heterogeneity Analysis Testing the Substitution Mechanism*

The framework in Section 4 posits that a biogas plant's success in fostering fertilizer substitution depends on the effective price ratio between chemical and organic inputs. Specifically, the optimal fertilizer portfolio is governed by the price differential: $\frac{w_\delta - w_\zeta}{w_\zeta}$. The incentive to substitute slurry for chemical fertilizer is stronger when the transaction costs of obtaining chemical fertilizer are higher, that is, when its effective price is greater. To test this mechanism directly, I examine whether treatment effects vary systematically

TABLE 5 *Heterogeneous Treatment Effects by Market Access*

	Dependent Variable: Monthly Avg. Turbidity (NTU)				
	(1)	(2)	(3)	(4)	(5)
Treatment (Baseline)	7.7 (12.5)	1.4 (18.3)	8.6 (21.4)	40.6* (22.4)	16.2*** (4.9)
Treatment × Distance to Market		1.7 (1.7)			14.0* (7.8)
Treatment × Distance to Fertilizer Shop			-0.12 (2.2)		-18.5** (9.1)
Treatment × Distance to Seed Center				-4.5 (4.3)	10.6** (4.4)
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes
Observations	27,033	27,033	27,033	27,033	27,033
R ²	0.39	0.39	0.39	0.39	0.39
Within R ²	0.0003	0.0004	0.0003	0.0004	0.0006

Notes: Standard errors clustered at the village level in parentheses. Distance measures are in kilometers (midpoints of original 1-5 scale: 1=0.5km, 2=1.5km, 3=3.5km, 4=7.5km, 5=15km). Column (1) includes only the baseline treatment effect. Columns (2)-(4) introduce each distance interaction separately. Column (5) includes all interactions simultaneously. *p<0.10, **p<0.05, ***p<0.01.

with village-level measures of market access, which proxy for the transaction costs of obtaining chemical fertilizers.

Table 5 presents results from five specifications that interact the treatment indicator with measures of distance to key agricultural inputs. The three distance measures capture distinct dimensions of market access. Distance to the fertilizer shop directly proxies the transaction cost of obtaining chemical fertilizer. The theoretical mechanism predicts that villages farther from fertilizer shops should experience larger turbidity reductions (more substitution away from chemical fertilizers) in response to a biogas plant, because the effective price of chemical alternatives is higher. Distance to market proxies for general economic activity and agricultural intensification, not specifically the price of chemical fertilizer. Villages farther from markets are typically more isolated, with potentially lower agricultural intensity and slower technology adoption, making the predicted sign ambiguous. Distance to seed center proxies for access to complementary agricultural inputs. Villages farther from seed centers may have less intensive agriculture overall,

potentially dampening the plant's impact, or distance to seed centers may correlate with general rural isolation.

Column (3) introduces the interaction between treatment and distance to fertilizer shops, the most direct test of the theoretical mechanism. The interaction coefficient is negative but very small in magnitude and statistically insignificant. This specification provides no evidence that villages farther from fertilizer shops experience larger turbidity reductions.

Column (4) introduces the interaction between treatment and distance to seed centers. The coefficient is negative but insignificant. The baseline treatment effect in this column is positive and marginally significant at the 10% level. This suggests that in villages at the minimum distance to seed centers, the biogas plant is associated with an *increase* in turbidity of approximately 41 NTU – a finding that is difficult to reconcile with the substitution hypothesis.

Column (5) includes all three interactions simultaneously. Here, a different pattern emerges. The negative coefficient on distance to fertilizer shops, now significant at the 5% level, is consistent with the theoretical mechanism: conditional on market and seed center access, villages farther from fertilizer shops experience greater turbidity reductions (i.e. more substitution away from chemical fertilizers) in response to a biogas plant. Specifically, a one-kilometer increase in distance to the nearest fertilizer shop is associated with an 18.5 NTU *larger reduction* in turbidity following plant installation. The positive interaction with distance to seed centers suggests that, conditional on other distances, villages farther from seed centers experience *smaller* turbidity reductions. This aligns with the interpretation that distance to seed centers proxies for general agricultural isolation and slower technology diffusion, which dampens the plant's effectiveness. The positive interaction with distance to market points in a similar direction, though the evidence is weaker.

Taken together, the heterogeneity results provide mixed support for the theoretical mechanism. The saturated model (column 5) yields a negative and significant interaction

with distance to fertilizer shops, consistent with the prediction that higher chemical fertilizer transaction costs strengthen substitution incentives. However, this result is not robust across specifications: it does not appear when the fertilizer shop interaction is entered alone (column 3). The positive and significant baseline treatment effect in column (5), indicating that plants *increase* turbidity in villages with the lowest distances to all markets, is difficult to explain under the substitution hypothesis. One possibility is that in well-connected villages, biogas plants facilitate agricultural intensification, or more fertilizer use overall, rather than substitution.

Thus, limitations to this specification are the following. The coefficients in column (5) should be interpreted with attention to the scale of the distance variables. The distance measures are categorical midpoints (1=0.5km, 2=1.5km, 3=3.5km, 4=7.5km, 5=15km). A one-unit increase in these scales corresponds to a substantially larger increment at higher distances. A one-standard-deviation increase in distance to fertilizer shops is associated with a treatment effect change of $(-18.5 \times SD)$ NTU.³ Given a baseline mean turbidity of approximately 65 NTU, this represents a modest reduction. Further, there is the concern that even with interactions, the treatment variables explain virtually none of the within-district, within-year, within-month variation in turbidity. The heterogeneity, while statistically detectable in some specifications, is not large enough to generate meaningful overall treatment effects. Lastly, the lack of robustness across columns (3) and (5) suggests that the significant result in column (5) may be sensitive to specification choices. Despite these caveats, the pattern of coefficients in Table 5 offers suggestive, though not definitive, evidence that the substitution mechanism theorized in Section 4 may be active in some contexts. The null average treatment effect documented in Section 7.2 masks meaningful heterogeneity, with some evidence that villages farther from fertilizer shops experience larger turbidity reductions. This finding, if confirmed in future work

³ The standard deviation of distance to fertilizer shops is 1.56 km, implying a one-standard-deviation increase is associated with a 28.9 NTU larger reduction in turbidity. This linear extrapolation should consider the concentration of the distance variable at the 7.5 km midpoint.

with more precise measures of fertilizer use and plant functionality, would reinforce the theoretical prediction that the plant's success hinges on local price conditions.

8. Conclusion

This study investigates whether community biogas plants installed under India's Gobar Dhan scheme can overcome transaction costs and create a functioning local manure market that induces substitution away from chemical fertilizers toward organic alternatives. The theoretical framework in Section 4 establishes that such plants have the potential to create welfare gains by converting low-efficiency dung into high-efficiency slurry, enabling both efficiency gains for mixed farms and gains-from-trade between specialized farms. However, the model also predicts that these gains depend critically on the relative price of chemical versus organic fertilizers and on local farming composition.

The empirical results reveal two main findings. First, there is no detectable average treatment effect of community biogas plants on water turbidity. Second, the heterogeneity analysis offers suggestive evidence that villages farther from fertilizer shops experience larger turbidity reductions in response to plant installation, consistent with the theoretical mechanism that substitution incentives are stronger where chemical fertilizer transaction costs are higher. However, these heterogeneity results are not robust across all specifications, and the within R^2 values indicate that treatment variables explain virtually none of the variation in turbidity.

The data limitations of this study point to directions for future research. The reliance on turbidity as a proxy for fertilizer use, necessitated by the absence of village-level fertilizer consumption data after 2018, introduces measurement error that biases estimates toward zero. The exploratory validation in Table 2 shows that turbidity explains little within-district variation in fertilizer use at the annual level, though this likely reflects aggregation bias rather than the absence of a relationship at the village level. Future work should measure fertilizer substitution directly using household-level data on fertilizer

expenditures and slurry uptake, or leverage remote sensing technologies that more directly capture nutrient application.

The findings also have several implications for policy design. The null average effect suggests that community biogas plants, as currently implemented, are not generating the hypothesized environmental benefits in the typical village. This does not mean the policy has failed; rather, it suggests that the conditions required for the substitution mechanism to operate may not be present in the average treated village. The heterogeneity results indicate that targeting plants to villages with poor access to chemical fertilizer markets – where the substitution incentive is strongest – could yield larger water quality improvements.

More broadly, the results suggest that the success of policies aimed at creating markets for organic inputs depends critically on local economic conditions, particularly the relative prices of chemical and organic alternatives. India's heavy fertilizer subsidies create a policy contradiction in which the government simultaneously subsidizes chemical fertilizers and promotes organic alternatives, making substitution economically irrational for farmers. The Government of India budgeted over Rs 1.67 lakh crore for fertilizer subsidies in 2025-26, representing nearly 70% of the total agriculture budget and approximately 40% of India's total subsidy spending (Indiaspend, 2025). The fertilizer subsidy allocation for 2024-25 climbed to Rs 1.91 lakh crore, driven by higher nutrient-based subsidies and volatile global import prices (BW Businessworld, 2025). Urea alone is sold to farmers at a statutorily notified maximum retail price of Rs 242 per 45 kg bag, even though the actual cost of production or import is approximately Rs 2,200 per bag, meaning the government absorbs over 85% of the true cost (PM India, 2023).

This massive subsidy creates several distortions with direct bearing on this study's findings. Critical to this study, the subsidy artificially depresses the price of chemical fertilizers relative to organic alternatives. Even when slurry is distributed free of charge, the effective price comparison is between chemicals whose price reflects massive government transfers and an organic alternative whose price reflects its true production

cost, minus any nominal distribution fee. As the Economic Survey 2025 has observed, reforms are urgently needed to correct unbalanced fertilizer use and reduce environmental externalities (Krishi Jagran, 2025).

Former Vice President Jagdeep Dhankhar, in office from 2022 to 2025, has publicly argued that if the total fertilizer subsidy of approximately Rs 3 lakh crore annually were handed directly to farmers rather than routed through manufacturers, farmers could decide for themselves whether to purchase chemical fertilizers or invest in organic alternatives such as cattle for manure (ANI, 2025). This direct transfer approach, which has been implemented in countries like the United States, would align farmer incentives with environmental goals by making the true cost of chemical fertilizers visible and allowing farmers to capture the savings from substitution.

Aligning price signals with environmental goals – either by reducing chemical fertilizer subsidies, increasing organic fertilizer subsidies, or both – would strengthen the incentive for farmers to substitute toward slurry. As long as chemical fertilizers remain artificially cheap relative to organic alternatives, even well-designed institutions to lower transaction costs will face an uphill battle. This policy implication suggests that lowering transaction costs through centralized processing is necessary but not sufficient. Complementary policies that align price signals – either through subsidy reform, targeted placement, or direct farmer payments – are also required for the environmental benefits of biogas infrastructure to materialize.

References

Aayog, Nity. (2023). "Production and Promotion of Organic and Bio Fertilisers with Special Focus on Improving Economic Viability of Gaushalas." The Task Force Report

Angelidaki, I., Treu, L., Tsapekos, P., Luo, G., Campanaro, S., Wenzel, H., & Kougias, P. G. (2018). Biogas upgrading and utilization: Current status and perspectives. *Biotechnology advances*, 36(2), 452–466. <https://doi.org/10.1016/j.biotechadv.2018.01.011>

ANI News. (2025, June 30). "If subsidies were handed directly to farmers, it could reshape Indian agriculture": VP Dhankhar. ANI News. <https://aninews.in/news/national/general-news/if-subsidies-were-handed-directly-to-farmers-it-could-reshape-indian-agriculture-vp-dhankhar20250630203809/>

Argus Media. (2024, February 2). "India to cut fertilizer budget by 13pc for 2024-25." <https://www.argusmedia.com/en/news-and-insights/latest-market-news/2533779-india-to-cut-fertilizer-budget-by-13pc-for-2024-25>

Argus Media. (2024, February 2). India to cut fertilizer budget by 13pc for 2024-25. Argus Media. <https://www.argusmedia.com/en/news-and-insights/latest-market-news/2533779-india-to-cut-fertilizer-budget-by-13pc-for-2024-25>

Asher, S., Lunt, T., Matsuura, R., & Novosad, P. (2019). The socioeconomic high-resolution rural-urban geographic dataset on India (SHRUG). URL: <https://doi.org/10.7910/DVN/DPESAK>.

Ashraf, M., and R. L. Christensen. "An Analysis of the Impact of Manure Disposal Regulations Farms." *Amer. J. Agr. Econ.* 56,2(1974):33 1-336.

Bardhan, P., & Udry, C. (1999). *Development Microeconomics*. Oxford University Press.

BW Businessworld. (2025, November). At a fertiliser crossroads, India weighs a profitable organic transition. BW Businessworld. <https://www.businessworld.in/article/at-a-fertiliser-crossroads-india-weighs-a-profitable-organic-transition-580997>

Carlin, N. T., Annamalai, K., Harman, W. L., & Sweeten, J. M. (2009). The economics of reburning with cattle manure-based biomass in existing coal-fired power plants for NO_x and CO₂ emissions control. *Biomass and Bioenergy*, 33(9), 1139-1157. <https://doi.org/10.1016/j.biombioe.2009.04.007>

Chandra, A., & Kaur, P. (2022). c, 160, 112681. <https://doi.org/10.1016/j.enpol.2021.112681>

Chanu, L. J., Kadu, P. P., & Singh, T. B. (2023). Integrated effect of phosphate solubilizing microorganisms and FYM levels on soil fertility, yield and nutrient uptake of soybean. *Indian Journal of Hill Farming*, *35*(Spcl).

Coase, R. H. (1960). The Problem of Social Cost. *The Journal of Law & Economics*, 3, 1–44. <https://doi.org/10.1086/466560>

Dandekar, Hemalata. (1983). "Gobar gas plants: how appropriate are they?" *Economic and Political Weekly*.

Foster, A. D., & Rosenzweig, M. R. (2010). Microeconomics of Technology Adoption. *Annual Review of Economics*, 2(1), 395–424. <https://doi.org/10.1146/annurev.economics.102308.124>

Ghosh, N. (2004). Reducing dependence on chemical fertilizers and its financial implications for farmers in India. *Ecological Economics*, 49(2), 149-162.

Ghosh, P., et al. (2020). Sustainability of community-scale biogas plants: A case study from Gujarat, India. *Journal of Cleaner Production*, 254, 120131. <https://doi.org/10.1016/j.jclepro.2020.120131>

Horan, R. D., & Shortle, J. S. (2005). When Two Wrongs Make a Right: Second-Best Point-Nonpoint Trading Ratios. *American Journal of Agricultural Economics*, 87(2), 340–352. <https://doi.org/10.1111/j.1467-8276.2005.00726.x>

Indiaspend. (2025, May). Why India's fertiliser subsidy bill is unsustainable. Indiaspend. <https://www.indiaspend.com/agriculture/why-indias-fertiliser-subsidy-bill-is-unsustainable-954386>

Innes, R. (2000). The economics of livestock waste and its regulation. *American Journal of Agricultural Economics*, 82(1), 97-117.

Kaur, J., Prusty, A. K., Ravisankar, N., Panwar, A. S., Shamim, M., Walia, S. S., Chatterjee, S., Pasha, M. L., Babu, S., Jat, M. L., López-Ridaura, S., Groot, J. C. J., Toorop, R. A., Barba-Escoto, L., Noopur, K., & Kashyap, P. (2021). Farm typology for planning targeted farming systems interventions for smallholders in Indo-Gangetic Plains of India. *Scientific Reports*, 11(1), 20978. <https://doi.org/10.1038/s41598-021-00372-w>

Khandker, Shahidur R.; Barnes, Douglas F.; Samad, Hussain A.(2012). Welfare Impacts of Rural Electrification: A Case Study from Bangladesh. *World Bank Policy Research Working Paper*, (6099). <https://doi.org/10.1596/1813-9450-4859>

Krishi Jagran. (2025, October). Hidden cracks policymakers must fix: A blueprint for sustainable and resilient Indian agriculture. Krishi Jagran. <https://krishijagran.com/blog/a-blueprint-for-sustainable-and-resilient-indian-agriculture/>

Kumar, S., Bhatt, B. P., Dey, A., Shivani, S., Kumar, U., Idris, M., Mishra, J. S., & Kumar, S. (2018). Integrated farming system in India: Current status, scope and future prospects in changing agricultural scenario. *The Indian Journal of Agricultural Sciences*, 88(11), 1661–1675. <https://doi.org/10.56093/ijas.v88i11.84880>

Lansing, S., Botero, R. B., & Martin, J. F. (2008). Waste treatment and biogas quality in small-scale agricultural digesters. *Bioresource technology*, 99(13), 5881-5890.

Li, Yanan, Mengru Wang, Xuanjing Chen, Shilei Cui, Nynke Hofstra, Carolien Kroeze, Lin Ma et al. "Multi-pollutant Assessment of River Pollution from Livestock Production Worldwide." *Water Research* 209, (2022): 117906. <https://doi.org/10.1016/j.watres.2021.117906>.

Liu, Y., Huang, T., Peng, D., Huang, J., Maurer, C., & Kranert, M. (2021). Optimizing the co-digestion supply chain of sewage sludge and food waste by the demand oriented biogas supplying mechanism. *Waste Management & Research: The Journal for a Sustainable Circular Economy*, 39(2), 302–313. <https://doi.org/10.1177/0734242X20953491>

Lu, J. (2013). The evolution and problems of China's biogas industry. *Chinese Journal of Population, Resources and Environment*, 11(2), 102–109. <https://doi.org/10.1080/10042857.2013.777200>

Lu, J., & Gao, X. (2021). Biogas: Potential, challenges, and perspectives in a changing China. *Biomass and Bioenergy*, 150, 106127.

Martin, G., Moraine, M., Ryschawy, J., Magne, M.-A., Asai, M., Sarthou, J.-P., Duru, M., & Therond, O. (2016). Crop–livestock integration beyond the farm level: A review. *Agronomy for Sustainable Development*, 36(3), 53. <https://doi.org/10.1007/s13593-016-0390-x>

Ministry of Drinking Water and Sanitation. (2018). Operational Guidelines: Galvanizing Organic Bio-Agro Resources Dhan (GOBAR-DHAN) Scheme. Government of India.

Ministry of New and Renewable Energy (MNRE). (2023). *Annual Report 2022-23*. Government of India.

Ministry of Statistics and Programme Implementation. (2019). Situation Assessment Survey of Agricultural Households: NSS 77th Round, January - December 201. National Sample Survey Office. <https://microdata.gov.in/NADA/index.php/catalog/157/related-materials>

Mint. (2025, June). Fertilizer subsidy bill faces West Asia shock. Mint. <https://www.livemint.com/news/india/fertilizer-subsidy-bill-west-asia-conflict-budget-fy26-11750681103512.html>

Moon, U.R., Durge, A.A., & Wadhai, V.S. (2024). Impacts of Agricultural Practices (Pesticides and Fertilizers) and Amalnala Stream on Amalnala Lake in Gadchandur, India. *Journal of Pure and Applied Microbiology*, 18(1), 389-400.

Mukherjee, S., Basak, A., Chakraborty, A., Goswami, R., Ray, K., Kumar, M., Kumar, S., ... & Verma, G. (2025). Effect of Organic and Inorganic Fertilizers on Crop Yield and Soil Fertility: A Comprehensive Review. *Journal of Experimental Agriculture International*, 47(2), 16-22.

Mwirigi, J., Balana, B. B., Mugisha, J., Walekhwa, P., Melamu, R., Nakami, S., & Makenzi, P. (2014). Socio-economic hurdles to widespread adoption of small-scale biogas digesters in Sub-Saharan Africa: A review. *biomass and bioenergy*, 70, 17-25.

National Biogas and Manure Management Programme (NBMMP). (2018). Guidelines for Implementation. Ministry of New and Renewable Energy, Govt. of India.

Pattanayak, S. K., Evan Mercer, D., Sills, E., & Yang, J. C. (2003). Taking stock of agroforestry adoption studies. *Agroforestry systems*, 57(3), 173-186.

Prashar, D., Singh, A., Dhama, V., Singh, P. K., Kumar, M., Kumar, S., ... & Verma, G. (2025). Effect of Organic and Inorganic Fertilizers on Crop Yield and Soil Fertility: A Comprehensive Review. *Journal of Experimental Agriculture International*, 47(2), 16-22.

Rathi, S. (2007). Optimization model for integrated municipal solid waste management in Mumbai, India. *Environment and development economics*, 12(1), 105-121.

Ribaudo, M., Kaplan, J. D., Christensen, L. A., Gollehon, N., Johansson, R., Breneman, V. E., ... & Peters, M. (2003). Manure management for water quality costs to animal feeding operations of applying manure nutrients to land. *USDA-ERS Agricultural Economic Report*, (824).

Sarma, V.V.S.S. et al. (2025). Do discharge of agricultural effluents and municipal sewage damage coastal ecosystem along the Indian coast? A reanalysis. *Journal of Earth System Science*, 134, Article 0036.

Schnitkey, G. D., & Miranda, M. J. (1993). The impact of pollution controls on livestock-crop producers. *Journal of Agricultural and Resource Economics*, 25-36.

Serdjuk, M., Bodmer, U., & Hülsbergen, K.-J. (2018). Integration of biogas production into organic arable farming systems: Crop yield response and economic effects. *Organic Agriculture*, 8(4), 301–314. <https://doi.org/10.1007/s13165-017-0196-4>

Shruthi, S. M., & Desai, N. R. M. (2021). Economic Analysis of Integrated Farming Systems in Uttar Kannada District of Karnataka. *Bhartiya Krishi Anusandhan Patrika*, Of. <https://doi.org/10.18805/BKAP342>

Stavins, R. N. (1995). Transaction Costs and Tradeable Permits. *Journal of Environmental Economics and Management*, 29(3), 133–148. <https://doi.org/10.1006/jeem.1995.1047>

Stonehouse, D. P., and A. V. S. Narayanan. "The Contributions of Livestock Manure to Profitability and to Self-Sufficiency in Plant Nutrients on Mixed Farms." *Can. J. Agr. Econ.* 32,1(1984):201-210.

Szarka, N., Scholwin, F., Trommler, M., Jacobi, H. F., Eichhorn, M., Ortwein, A., & Thrän, D. (2013). A novel role for bioenergy: a flexible, demand-oriented power supply. *Energy*, 61, 18-26.

Weiland P. (2010). Biogas production: current state and perspectives. *Applied microbiology and biotechnology*, 85(4), 849–860. <https://doi.org/10.1007/s00253-009-2246-7>

Wright, I. A., Tarawali, S., Blümmel, M., Gerard, B., Teufel, N., & Herrero, M. (2012). Integrating crops and livestock in subtropical agricultural systems. *Journal of the Science of Food and Agriculture*, 92(5), 1010–1015. <https://doi.org/10.1002/jsfa.4556>

Yalcinkaya, S. (2020) A spatial modeling approach for siting, sizing and economic assessment of centralized biogas plants in organic waste management. *J. Clean. Prod.*, 255.

Yap, C, K. Foster, P. Preckel, O. Doering, and B. Richert. "Mitigating the Compliance Cost of a Phosphorus-Based Swine Manure Management Policy." *J. Agr. and Appl. Econ.* 36,1(2004):23-34.

Zhou, Y., N. Jain, G. K. Jha, and T. Begho. (2022)."Nitrogen Pollution from Cattle Production in India: A Review of the Social, Cultural and Economic Influences." *The Journal of Agricultural Science* 160, no. 1–2: 98–106. <https://doi.org/10.1017/S0021859622000120>.